

Escola Platform - Database Schema

Schemas	Tables	Lookup Tables	Extensions
6	41	19	uuid-oss, pgcrypto, unaccent

IDENTITY SCHEMA

Table: tenants
tenant_id (PK)
tenant_uuid (UK)
tenant_type_id (FK)
tenant_name
tenant_name_normalized
tenant_config (JSONB)
is_active
+ audit fields

Table: tenant_types [LOOKUP]
type_id (PK)
type_uuid (UK)
tenant_id (FK)
type_code (UK)
type_name
description
max_students
max_teachers
max_storage_gb
has_analytics
has_api_access
has_white_label
pricing_tier
icon_name
color_code
display_order

ASSETS SCHEMA

Table: app_users
user_id (PK)
user_uuid (UK)
tenant_id (FK)
auth_user_id (UK)
full_name
full_name_normalized
email
email_normalized
user_role_id (FK)
is_active
user_config (JSONB)
+ audit fields

Table: user_roles [LOOKUP]
role_id (PK)
role_uuid (UK)
tenant_id (FK)
role_code (UK)
role_name
description
permission_level
can_manage_students
can_manage_content
can_manage_users
can_manage_tenant
icon_name
color_code
display_order

Table: media_files
file_id (PK)
file_uuid (UK)
tenant_id (FK)
mime_type_id (FK)

storage_path
original_name
original_name_normalized
size_bytes
alt_text
alt_text_normalized
metadata (JSONB)
+ audit fields

Table: mime_types [LOOKUP]
mime_type_id (PK)
mime_type_uuid (UK)
tenant_id (FK)
category_id (FK)
mime_type_code (UK)
mime_type_name
file_extension
max_file_size_mb
icon_name
display_order

Table: media_categories [LOOKUP]
category_id (PK)
category_uuid (UK)
tenant_id (FK)
category_code (UK)
category_name
description
default_icon_name
default_max_size_mb
display_order

Table: students
student_id (PK)
student_uuid (UK)

SCHOOL SCHEMA

tenant_id (FK)
user_id (FK, Nullable - Hybrid Auth)
nickname + normalized
first_name + normalized
middle_name + normalized
last_name + normalized
birth_date
avatar_config (JSONB)
is_active
+ audit fields

Table: guardians
guardian_id (PK)
guardian_uuid (UK)
tenant_id (FK)
user_id (FK)
phone_number
relationship_type_id (FK)
is_primary
+ audit fields

Table: relationship_types [LOOKUP]
relationship_type_id (PK)
relationship_uuid (UK)
tenant_id (FK)
relationship_code (UK)
relationship_name
description
can_authorize
requires_legal_proof
icon_name
display_order

Table: student_guardians [JUNCTION]
student_guardian_id (PK)
student_id (FK)
guardian_id (FK)
tenant_id (FK)

relationship_notes + normalized
is_authorized_pickup
+ audit fields

Table: classes
class_id (PK)
class_uuid (UK)
tenant_id (FK)
class_name + normalized
grade_level_id (FK)
school_year_id (FK)
max_students
class_config (JSONB)
is_active
+ audit fields

Table: grade_levels [LOOKUP]
grade_level_id (PK)
grade_uuid (UK)
tenant_id (FK)
grade_code (UK)
grade_name
description
age_range_min
age_range_max
display_order
icon_name
color_code

Table: school_years [LOOKUP]
school_year_id (PK)
year_uuid (UK)
tenant_id (FK)
year_code (UK)
year_name
description
start_date
end_date

is_current
display_order

Table: teachers
teacher_id (PK)
teacher_uuid (UK)
tenant_id (FK)
user_id (FK)
specialization_id (FK)
hire_date
is_active
+ audit fields

Table: specializations [LOOKUP]
specialization_id (PK)
specialization_uuid (UK)
specialization_name (UK)
description
is_active
+ audit fields

Table: certifications [LOOKUP]
certification_id (PK)
certification_uuid (UK)
certification_name (UK)
description
issuing_organization
is_active
+ audit fields

Table: teacher_certifications [JUNCTION]
teacher_id (FK)
certification_id (FK)
obtained_date
expiry_date
credential_number
notes
+ audit fields

Table: subjects [LOOKUP]
subject_id (PK)
subject_uuid (UK)
subject_name (UK)
description
grade_level_id (FK)
is_active
+ audit fields

Table: teacher_subjects [JUNCTION]
teacher_id (FK)
subject_id (FK)
proficiency_level_id (FK)
years_experience
is_primary_subject
notes
+ audit fields

Table: proficiency_levels [LOOKUP]
proficiency_level_id (PK)
proficiency_uuid (UK)
tenant_id (FK)
proficiency_code (UK)
proficiency_name
description
minimum_years
icon_name
color_code
display_order

Table: class_students [JUNCTION]
class_student_id (PK)
class_id (FK)
student_id (FK)
tenant_id (FK)
enrollment_date
status_id (FK)

+ audit fields

Table: enrollment_statuses [LOOKUP]
status_id (PK)
status_uuid (UK)
tenant_id (FK)
status_code (UK)
status_name
description
allows_attendance
allows_grading
is_final_state
icon_name
color_code
display_order

CONTENT SCHEMA

Table: modules
module_id (PK)
module_uuid (UK)
module_name + normalized
description + normalized
module_type_id (FK)
difficulty_level
recommended_age_min/max
display_order
thumbnail_url
module_config (JSONB)
is_published
+ audit fields

Table: module_types [LOOKUP]
module_type_id (PK)
module_type_uuid (UK)
tenant_id (FK)
module_type_code (UK)

module_type_name
description
icon_name
color_code
display_order

Table: activities
activity_id (PK)
activity_uuid (UK)
module_id (FK)
activity_name + normalized
description + normalized
activity_type_id (FK)
display_order
estimated_duration
points_reward
activity_data (JSONB)
thumbnail_url
is_published
+ audit fields

Table: activity_types [LOOKUP]
activity_type_id (PK)
activity_type_uuid (UK)
tenant_id (FK)
activity_type_code (UK)
activity_type_name
description
icon_name
default_points
requires_interaction
display_order

Table: activity_resources
resource_id (PK)
resource_uuid (UK)
activity_id (FK)
resource_name + normalized

resource_type_id (FK)
media_file_id (FK)
display_order
is_required
usage_context_id (FK)
resource_config (JSONB)
is_published
+ audit fields

Table: resource_types [LOOKUP]
resource_type_id (PK)
resource_type_uuid (UK)
tenant_id (FK)
resource_type_code (UK)
resource_type_name
description
mime_types (JSONB)
max_file_size_mb
icon_name
display_order

Table: usage_contexts [LOOKUP]
usage_context_id (PK)
usage_context_uuid (UK)
tenant_id (FK)
context_code (UK)
context_name
description
icon_name
display_order

Table: student_progress
progress_id (PK)
progress_uuid (UK)
student_id (FK)

GAME SCHEMA

activity_id (FK)
tenant_id (FK)
status_id (FK)
score
attempts_count
time_spent_seconds
progress_data (JSONB)
completion_date
+ audit fields

Table: progress_statuses [LOOKUP]
status_id (PK)
status_uuid (UK)
tenant_id (FK)
status_code (UK)
status_name
description
icon_name
color_code
is_final_state
display_order

Table: badges
badge_id (PK)
badge_uuid (UK)
badge_name + normalized
description + normalized
badge_type_id (FK)
icon_url
rarity_id (FK)
points_value
unlock_criteria (JSONB)
display_order
is_active
+ audit fields

Table: badge_types [LOOKUP]
badge_type_id (PK)

badge_type_uuid (UK)
badge_type_code (UK)
badge_type_name
description
icon_name
display_order
is_active
+ audit fields

Table: badge_rarities [LOOKUP]
rarity_id (PK)
rarity_uuid (UK)
rarity_code (UK)
rarity_name
description
color_code
points_multiplier
display_order
is_active
+ audit fields

Table: student_badges
student_badge_id (PK)
student_id (FK)
badge_id (FK)
tenant_id (FK)
earned_at
earn_metadata (JSONB)
+ audit fields

Key Relationships

From	To	Type	Description
tenants	tenant_types	N:1	Tenant classification (school/enterprise)
tenants	app_users	1:N	Multi-tenant user isolation
app_users	user_roles	N:1	User role with permissions
tenants	students	1:N	School students
app_users	students	0..1:1	Optional student login (Hybrid Auth Model)
tenants	media_files	1:N	Tenant media library
media_files	mime_types	N:1	File type classification
mime_types	media_categories	N:1	MIME category
app_users	guardians	1:1	User is guardian
app_users	teachers	1:1	User is teacher
guardians	relationship_types	N:1	Guardian relationship type
students	student_guardians	1:N	Student has guardians
guardians	student_guardians	1:N	Guardian has students
students	class_students	1:N	Student enrollment
classes	class_students	1:N	Class roster
class_students	enrollment_statuses	N:1	Enrollment status
classes	grade_levels	N:1	Class grade level
classes	school_years	N:1	Academic year
teachers	specializations	N:1	Teacher specialization
teachers	teacher_certifications	1:N	Teacher certifications
certifications	teacher_certifications	1:N	Certification type
teachers	teacher_subjects	1:N	Subjects taught
subjects	teacher_subjects	1:N	Teachers for subject
teacher_subjects	proficiency_levels	N:1	Teaching proficiency
subjects	grade_levels	N:1	Subject grade level
modules	module_types	N:1	Module classification
modules	activities	1:N	Module contains activities

activities	activity_types	N:1	Activity classification
activities	activity_resources	1:N	Activity uses resources
activity_resources	resource_types	N:1	Resource classification
activity_resources	usage_contexts	N:1	Resource usage context
activity_resources	media_files	N:1	Media library reference
students	student_progress	1:N	Progress tracking
activities	student_progress	1:N	Activity completion
student_progress	progress_statuses	N:1	Progress status
students	student_badges	1:N	Earned badges
badges	student_badges	1:N	Badge awards
badges	badge_types	N:1	Badge classification
badges	badge_rarities	N:1	Badge rarity level

Key Features

Feature	Description
Hybrid ID Strategy	Internal INT (student_id) + External UUID (student_uuid)
3NF Normalization	All free-text enums converted to lookup tables with metadata
Text Normalization	30+ *_normalized columns for accent-insensitive search (José → jose)
Multi-Tenancy	RLS enabled for tenant isolation via tenant_id
Audit Trail	All tables: created_at/by, updated_at/by, deleted_at (soft delete)
JSONB Validation	CHECK constraints ensure data integrity for all JSONB columns
Lookup Tables	19 lookup tables with icons, colors, metadata for rich UI
Extensions	uuid-osspp (UUID gen), pgcrypto (encryption), unaccent (search)